

Mohammad Javad Hoseini

SOFTWARE / SITE RELIABILITY ENGINEER

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🔗 vector-mj | 💼 Mohammad Javad Hoseini | 📁 Portfolio

“Hope is not a strategy”

Experience

ArvanCloud

Tehran

SOFTWARE ENGINEER (CONTRACT)

Nov 2025 - May 2026

- Architected and developed a **Scalable, Event-driven, MVCS** REST API with **FastAPI** and **gRPC** communications; the MAV (Multi-Antivirus, like VirusTotal) system parallelizes file scans across **6 engines** (ESET, Kaspersky, ClamAV, Dr.Web, Padvish, YARA), scanning files ≤ 10 MB in **~2.75 s** on average and sustaining **~1.9M scans/day**.
- Leveraged **Message Brokers (RabbitMQ/Kafka + Schema Registry)** to decouple file ingestion from scanning services, ensuring system resilience during high-traffic bursts.
- Inter-service communication (IPC) based on **gRPC** and **Protobuf** for the whole scan cycle and monitoring sub-service heartbeats, reducing terabyte-scale Kafka topics by **~90%** to **150 GB**.

OutFlow (Startup)

Khorasan Razavi, Mashhad

SOFTWARE ENGINEER / TEAM LEAD

Jan 2025 - Nov 2025

- Led a **3-member engineering team**, establishing a scrum-lite framework with bi-weekly syncs to unblock development and maintain project velocity.
- Architected a high-performance **MVCS REST API** using **FastAPI** and **Neo4j**, implementing **JWT authentication / session management, TOTP** and a graph-based data model to handle complex learning-path relationships.
- Developed the frontend with **Vue.js, D3.js, Typescript, Pinia**, and CSS (Flex, Grid and Sass).

ArvanCloud

Tehran - Hybrid

SITE RELIABILITY ENGINEER

June 2022 - Dec 2024

MIDDLEWARE TEAM (Nov 2023 - Dec 2024)

- Implemented Chaos Engineering practices using **Chaos Mesh** to orchestrate fault injection experiments (network latency, pod failures, and I/O stress), identifying and remediating single points of failure in service inter-dependencies.
- Engineered a global latency monitoring system with **Python/FastAPI** to track API response times across 45+ cities in 28 countries, reducing tail latencies.
- Wrote custom **Ansible** modules (built in **2 weeks**, replacing a planned KMS) that rotate credentials across all **18 database instances** (8 MongoDB, 8 Redis, 1 ClickHouse, 1 CockroachDB) for the whole team in **under 5 minutes**.
- Ran **8 Game Day** exercises on critical notification, payment, and authentication services, driving **13 new alerting rules** and **5 dashboards**.

OBJECT STORAGE TEAM (JUNE 2022 - NOV 2023)

- Automated Ceph cluster operations by **Pythonic Ansible** modules across **90+ servers**, **reducing human error** and eliminating partial service outages during maintenance.
- Architected a metadata synchronization pipeline** between a **~5 PB** Ceph cluster and **Elasticsearch** to enable advanced user analytics and Multi-AV services.
- Designed and managed a **4-SRE 24/7 on-call rotation (1-2 incidents/month, ~4 hr MTTR** for minor incidents), and standardized staging with **Infrastructure as Code (Ansible)** for dev/production parity.

DigiEdge - On site

DevOps Engineer

Khorasan Razavi, Mashhad

Sep 2021 - Mar 2022

- Created CI/CD pipelines for Android and iOS developers with **Microsoft Azure**.
- Developed a customized **MTPROTO** proxy server in **Javascript** with per-user limits on time, data usage, and connection count.

Skills

Programming	Python, Javascript/Typescript, Lua, Bash
Web & Frameworks	HAProxy, FastAPI, Flask, Express.js
Databases	MongoDB, Redis, Elasticsearch, Memgraph, Neo4j, Clickhouse, CockroachDB, MySQL
Cloud & Orchestration	Kubernetes, Helm, Docker Swarm, Ansible, Terraform, Vault
CI/CD & Observability	Git/GitLab, Jaeger, Sentry, Prometheus, Grafana
Stream/Message brokers	Kafka, RabbitMQ
Networking & OS	Hands-on: Debian-based Linux (LPIC-2), TCP/IP, VRRP, iptables, Tunneling. Theory/Lab: BSD, BGP, MPLS.
Systems Internals & Security (Hands-on/Lab)	eBPF (Aya/Tetragon), LSM (AppArmor, SELinux), OPA
Languages	Persian (Native), English (Intermediate)

Education

Sajad University of Technology

B.S. in Computer Engineering

Razavi Khorasan, Mashhad

Jan. 2026

Community & Impact

Censorship Circumvention Infrastructure

During Iran internet outages - 2026

- Deployed and managed a proxy cluster (based on **Xray**, **Tor** transport modules and tunneling methods) providing unrestricted internet access for **150+** users (**~1 TB/month**) **on a personal budget**.

SRE Article

Aug. 2025

- Authored *"A review of increasing site stability with Site Reliability Engineering (SRE) and Chaos Engineering"* exploring the strategic integration of fault injection to reduce MTTR.